Multivariate Analysis of Air Pollution Patterns Across Monitoring Stations in India



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by

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Declaration

I, **Soham Choudhury**, hereby declare that the project titled ***Multivariate Analysis of Air Pollution Patterns Across Monitoring Stations in India*** and the work presented therein are my own and original, carried out under the supervision of **Prof. Sanjeev Kumar, Department of Statistics**, Banaras Hindu University, during the period from January 2026 to May 2026. This project has been submitted to the DST-Centre for Interdisciplinary Mathematical Sciences, Banaras Hindu University, in partial fulfilment of the requirements for the M.Sc. in Statistics & Computing course. To the best of my knowledge and belief, the information and data presented in this project are authentic. This work has not been submitted to any other university or institution for the award of any degree, diploma, or fellowship, nor has it been published previously.

Place: Varanasi, India **Soham Choudhury**

Date: 6th May, 2026 **24419STC009**

CERTIFICATE

This is to certify that **Soham Choudhury**, a student of the **Master of Science in Statistics & Computing**, Semester IV of the **DST-Centre for Interdisciplinary Mathematical Sciences**, Institute of Science, Banaras Hindu University, bearing exam roll no. **24419STC009** and enrollment no. **482755** of 2024-2026 has successfully completed his project entitled "**Multivariate Analysis of Air Pollution Patterns Across Monitoring Stations in India**". The project was completed under my direct supervision and guidance for the partial fulfillment of the course curriculum of Master of Science in Statistics & Computing. He carried out the whole work with utmost sincerity, and the present report is the outcome of his sincere efforts and endeavors. This work either in the present form or in part in any other form is not being submitted anywhere to the best of my knowledge.

Supervisor

Course Coordinator

Coordinator, DST-CIMS

Acknowledgment

I am honoured to have this opportunity to convey my sincere gratitude to my supervisor, **Prof. Sanjeev Kumar**, for his precious supervision, constant motivation, and caring mentorship throughout the tenure of this research. Without his immense knowledge and unparalleled vision, this thesis would not have taken this shape.

Place: Varanasi, India **Soham Choudhury**

Date: 6th May, 2026 **24419STC009**

Dedicated to,

My parents and teachers, whose guidance and support made this work possible.

Abstract

Air pollution is a complex environmental issue influenced by multiple interacting sources, making it difficult to analyze using conventional approaches. This study explores the spatial structure of air pollution across monitoring stations using multivariate statistical techniques. Data on major pollutants, including $PM2.5$, $PM10$, NO_2, SO_2, CO, $Ozone$, and NH_3, were analyzed after appropriate preprocessing and transformation.

Principal Component Analysis (PCA) was first applied to assess the underlying dimensionality of the data, indicating the presence of multiple contributing factors. Factor Analysis (FA) was subsequently used to identify interpretable pollution sources, corresponding to particulate, vehicular and secondary, and industrial emission sources.

To further understand how these pollution patterns manifest across monitoring stations, K-means clustering was applied to the factor scores. The analysis identified three distinct pollution regimes: a low pollution group, a mixed industrial–vehicular group, and an urban particulate-dominated group. Although the cluster separation was moderate, this reflects the continuous and overlapping nature of pollution sources rather than limitations of the methodology.

Overall, the findings highlight that air pollution is multidimensional and varies significantly across locations, with no sharply defined boundaries between pollution types. These results emphasize the importance of region-specific and multi-source strategies for effective air quality management.

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1 Introduction

1.1 Background

Air pollution poses an urgent and escalating threat to both environmental and public health worldwide, particularly in rapidly developing countries such as India. Rapid urbanization, industrial growth, and increasing vehicular emissions have significantly contributed to the deterioration of air quality. Pollutants such as Particulate Matter ($PM2.5$ and $PM10$), nitrogen dioxide (NO_2), Sulfur Dioxide (SO_2), Carbon Monoxide (CO), Ozone, and Ammonia (NH_3) are widely recognized for their adverse effects on human health and the environment.

Air pollution is inherently a multidimensional phenomenon, where multiple pollutants interact and originate from diverse sources such as traffic, industries, and atmospheric chemical processes. As a result, analyzing pollutants individually often fails to capture the underlying relationships and combined effects that define the overall pollution structure. Understanding these interactions is essential for effective monitoring and policy-making.

1.2 Motivation

Traditional approaches to air pollution analysis tend to focus on individual pollutants or simple summary statistics, which may not adequately represent the complexity of pollution patterns. In reality, pollution at a given location is influenced by multiple sources simultaneously, leading to overlapping and interdependent effects.

This complexity motivates the use of multivariate statistical techniques that can analyze multiple variables together and uncover hidden structures within the data. Techniques such as Principal Component Analysis (PCA) and Factor Analysis (FA) allow for dimensionality reduction and identification of latent pollution sources, while clustering methods help in grouping monitoring stations with similar pollution characteristics.

By applying these methods, the study aims to move beyond surface-level analysis and provide a deeper understanding of how pollution behaves across different regions. Such insights are valuable for designing targeted and effective air quality management strategies.

1.3 Objectives

The primary objectives of this study are:

1. To analyze the distribution and variability of major air pollutants across monitoring stations.

2. To identify underlying pollution factors using multivariate statistical techniques such as PCA and Factor Analysis

3. To classify monitoring stations into groups based on their pollution profiles using clustering methods.

1.4 Structure of the Project

The remaining parts of this report are organized as follows:

- **Chapter 2: Literature Review** presents an overview of previous studies related to air pollution analysis and the use of multivariate techniques.
- **Chapter 3: Data Description and Preprocessing** describes the dataset, variables considered, and the preprocessing steps applied, including data cleaning and transformation.
- **Chapter 4: Methodology** explains the statistical methods used in the study, including PCA, Factor Analysis, and clustering techniques.
- **Chapter 5: Results** presents the findings from the analysis and provides interpretation of results.
- **Chapter 6: Conclusion** summarizes the key insights and discusses possible directions for future research.

2 Literature Review

Air pollution has been widely studied due to its significant impact on environmental quality and public health. Over time, researchers have increasingly relied on statistical and multivariate techniques to analyze pollution data, as these methods are capable of handling multiple interrelated variables simultaneously. This section reviews key studies related to air pollution analysis and the application of multivariate techniques such as Principal Component Analysis (PCA), Factor Analysis (FA), and clustering methods.

2.1 Studies on Air Pollution Analysis

Existing literature establishes air pollution as a complex phenomenon influenced by multiple sources, including vehicular emissions, industrial activities, and atmospheric processes. Several studies have highlighted significant spatial and temporal variations in pollutant concentrations across different regions (World Health Organization, 2016). In developing countries like India, rapid urbanization and industrialization have led to increased pollution levels, particularly in urban areas.

Research has shown that pollutants such as particulate matter ($PM2.5$ and $PM10$) are often associated with traffic emissions and dust, while gases like sulfur dioxide (SO_2) are linked to industrial activities (Guttikunda & Goel, 2013). These studies emphasize that pollution cannot be adequately understood by analyzing individual pollutants in isolation, as multiple sources often contribute simultaneously.

2.2 Application of Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is widely used in environmental studies to reduce the dimensionality of datasets and identify dominant patterns. PCA transforms correlated variables into a reduced set of uncorrelated components that capture the maximum variance in the data (Jolliffe, 2002).

In air pollution studies, PCA has been applied to identify major contributing factors and simplify complex datasets involving multiple pollutants. For example, Viana et al. (2008) used PCA to analyze particulate matter sources and found that a few principal components could explain a significant proportion of the variability. However, while PCA is effective in capturing variance, its components are often difficult to interpret in terms of real-world pollution sources.

2.3 Factor Analysis for Source Identification

Factor Analysis (FA) provides a model-based approach to identify latent variables underlying observed data. Unlike PCA, FA focuses on explaining the correlations among variables through a smaller number of underlying factors (Harman, 1976).

FA has been widely used in environmental studies particularly for source apportionment. For instance, Thurston and Spengler (1985) applied FA to air quality data and successfully identified distinct pollution sources such as traffic emissions, industrial contributions, and secondary atmospheric processes. Similar applications have shown that particulate matter is often associated with combustion and dust sources, nitrogen oxides with vehicular emissions, and sulfur dioxide with industrial activities. Due to its interpretability, FA is particularly useful for understanding the underlying structure of pollution data and linking statistical results to real-world phenomena.

2.4 Clustering Techniques in Air Pollution Studies

Clustering methods are commonly used to group monitoring locations based on similarities in pollution characteristics. K-means clustering, introduced by MacQueen (1967), is one of the most widely used methods due to its simplicity and efficiency. In environmental studies, clustering has been applied to classify regions into pollution categories such as high, moderate, and low pollution zones. For example, Jain (2010) discussed the effectiveness of clustering techniques in identifying patterns in large datasets, including environmental data.

However, several studies have noted that pollution data often exhibit overlapping patterns due to the presence of multiple sources. As a result, clusters may not be sharply defined, and moderate clustering performance is common. This reflects the continuous nature of environmental data rather than limitations of the clustering methods.

2.5 Summary of Literature

The literature indicates that air pollution is a multidimensional phenomenon influenced by multiple interacting factors. Multivariate techniques such as PCA and Factor Analysis are effective in identifying underlying structures and pollution sources, while clustering methods provide a way to group locations based on these patterns. Despite the usefulness of these methods, many studies emphasize that pollution profiles often overlap, making it difficult to delineate clearly distinct pollution categories. This highlights the importance of using a combination of statistical techniques to gain a comprehensive understanding of pollution patterns.

3 Data Description and Preprocessing

3.1 Data Source

The dataset used in this study consists of air pollution measurements collected from multiple monitoring stations across different states in India. The data includes observations for major air pollutants that are commonly used to assess air quality.
The source of the dataset is: https://www.data.gov.in/resource/real-time-air-quality-index-various-locations

3.2 Variables Considered

The analysis focuses on the following pollutants:

- $PM2.5$ - Fine particulate matter (particles $\leq 2.5\ \mu m$)
- $PM10$ - Coarse particulate matter (particles $\leq 10\ \mu m$)
- NO_2 - Nitrogen dioxide
- SO_2 - Sulfur dioxide
- CO - Carbon monoxide
- $Ozone$ - Ground-level ozone
- NH_3 - Ammonia

The average pollution concentration of these pollutants are taken into consideration.
In addition to pollutant concentrations, the dataset also contains:

- Station name
- State
- City
- Geographic coordinates (latitude and longitude)

3.3 Data Cleaning and Preprocessing

The raw dataset contained missing values and non-numeric entries such as "NA", "–", and blank values. These were first converted into standard missing values (NA) to ensure consistency in the analysis. Duplicate observations for the same station and pollutant were identified and handled

by averaging the values, ensuring that each station-pollutant pair had a unique representative value. The dataset was then reshaped from long to wide format, where each row corresponds to a monitoring station and each column represents a pollutant.

3.4 Handling Missing Values

Missing values were present across several pollutants. To ensure reliability of multivariate analysis, only those stations with complete observations across all selected pollutants were retained. This resulted in a reduced but consistent dataset suitable for further analysis.

3.5 Data Transformation and Scaling

Air pollution data exhibited skewed distributions due to the presence of extreme values. To address this, a logarithmic transformation $\log(1 + x)$ was applied to pollutant concentrations.

Following transformation, the data were standardized to have zero mean and unit variance. This step ensures that all variables contribute equally to the analysis and prevents variables with larger scales from dominating the results.

3.6 Final Dataset

After preprocessing, the final dataset consists of monitoring stations with complete observations for all pollutants, transformed and standardized for use in multivariate analysis techniques such as PCA, Factor Analysis and clustering.

4 Methodology

4.1 Overview

This study employs a combination of multivariate statistical techniques to analyze the structure of air pollution data. The methodology consists of three main steps:

1. Exploratory structure analysis using Principal Component Analysis (PCA).
2. Identification of latent pollution factors using Factor Analysis (FA).
3. Grouping of monitoring stations using K-means clustering.

Additionally, non-parametric statistical tests are used to assess differences in pollution levels across states.

4.2 Kruskal-Wallis Test

The Kruskal-Wallis Test is a non-parametric statistical method used to determine whether there are significant differences between two or more independent groups. It is an extension of the Mann-Whitney U test and serves as a non-parametric alternative to one-way ANOVA when the normality assumptions are violated.

In this study, the Kruskal-Wallis test is used to examine whether the distribution of pollutant concentrations differs across states.

4.2.1 Hypotheses

The hypotheses for Kruskal-Wallis test are defined as follows:

- **Null Hypothesis(H_0):** The distribution of pollutant concentrations is the same across all states.
- **Alternative Hypothesis(H_1):** At least one state has a different distribution of pollutant concentrations.

4.2.2 Test Statistic

The Kruskal-Wallis test statistic is given by:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} n_i (\bar{R}_i - \bar{R})^2$$

where:

- N is the total number of observations
- k is the number of groups (states)
- n_i is the number of observations in the i-th group
- $\bar{R}_i$ is the average rank of i-th group
- $\bar{R} = \frac{N+1}{2}$ is the overall average rank

4.2.3 Distribution of the Test Statistic

Under the null hypothesis, the test statistic approximately follows a chi-square distribution:

$$H \sim \chi^2_{k-1}$$

where $k-1$ represents the degrees of freedom

4.2.4 Decision Rule

The null hypothesis is rejected if $H_{obs} > \chi^2_{\alpha,k-1}$. Also the null hypothesis is rejected if the p-value associated with the test statistic is less than the chosen level of significance α (typically 0.05). Rejection of the null hypothesis indicates that there is a statistically significant difference in pollutant concentrations across at least one pair of states.

4.3 Principal Component Analysis(PCA)

Principal Component Analysis (PCA) is a multivariate statistical technique used to examine the structure of a dataset by transforming a set of correlated variables into a smaller set of uncorrelated variables called principal components. These components are linear combinations of the original variables and are constructed to capture the maximum possible variance in the data.

In this study, PCA is used as an exploratory tool to understand the underlying structure of air pollution data and to assess the number of significant dimensions present in the dataset.

4.3.1 Objective of PCA

The main objectives of applying PCA in this study are:

- To identify the dominant patterns of variation in pollutant concentrations
- To determine the number of significant components
- To examine relationships among different pollutants

4.3.2 Mathematical Formulation

Let $X = (X_1, X_2, \cdots, X_p)^T$ represent a vector of standardized variables. The first principal component is defined as:

$$PC_1 = a_{11}X_1 + a_{12}X_2 + \cdots + a_{1p}X_p$$

subject to the constraint:

$$\sum_{i=1}^{p} a_{1j}^2 = 1$$

The coefficients a_{1j} are chosen such that the variance of PC_1 is maximized. Subsequent components are defined similarly, with the additional constraint that they are orthogonal (uncorrelated) to the previous components.

4.3.3 Eigenvalue Decomposition

PCA is based on the eigenvalue decomposition of the covariance (or correlation) matrix Σ:

$$\Sigma a_i = \lambda_i a_i$$

where:

- λ_i represents the eigenvalue corresponding to the i-th principal component.
- a_i is the eigenvector defining the direction of the component

The eigenvalues indicate the amount of variance explained by each component. Scree plot and how much variance explained by the components is used to determine the number of principal components.

4.3.4 Interpretation

Each principal component represents a combination of pollutants that explains a portion of the total variance. Components with higher eigenvalues explain more variation and are considered more important.

In this study, PCA is not used for dimensionality reduction in subsequent analysis but serves as an exploratory step to guide the application of Factor Analysis.

4.4 Factor Suitability Testing

Before applying Factor Analysis, it is necessary to assess whether the dataset is suitable for such analysis. This is done using the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity.

4.4.1 Kaiser-Meyer-Olkin (KMO) Test

The KMO test measures the adequacy of the data for Factor Analysis by comparing the magnitude of observed correlation coefficients with that of partial correlations.

The KMO statistic is defined as:

$$KMO = \frac{\sum_{j \neq k} r_{jk}^2}{\sum_{j \neq k} r_{jk}^2 + \sum_{j \neq k} p_{ij}^2}$$

where r_{jk} are observed correlations and p_{jk} are the partial correlations. A KMO value above 0.5 is considered the minimum acceptable threshold for Factor Analysis.

4.4.2 Bartlett's Test of Sphericity

Bartlett's test examines whether the correlation matrix is significantly different from an identity matrix.

Here:

- **Null Hypothesis(H_0):** The correlation matrix is an identity matrix
- **Alternative Hypothesis(H_1):** The correlation matrix is not an identity matrix

The test statistic is given by:

$$\chi^2 = -(n - 1 - \frac{2p + 5}{6}) ln|R|$$

where:

- n = number of observations
- p = number of variables
- $|R|$ = determinant of the correlation matrix

If H_0 is rejected then we can proceed with Factor Analysis

4.5 Factor Analysis (FA)

Factor Analysis is a multivariate statistical technique used to identify underlying latent variables (factors) that explain the correlations among observed variables. Unlike Principal Component Analysis, which focuses on variance, Factor Analysis provides a model-based approach that assumes observed variables are influenced by a smaller number of unobserved factors.

In this study, Factor Analysis is used to identify underlying pollution sources that govern the observed pollutant concentrations.

4.5.1 Objectives of Factor Analysis

The main objectives of applying Factor Analysis in this study are:

- To identify latent pollution factors influencing observed variables
- To reduce the dimensionality of the dataset while preserving interpretability
- To group pollutants based on common underlying sources

4.5.2 Mathematical Model

The Factor Analysis model assumes that each observed variable can be expressed as a linear combination of common factors and a unique error term:

$$X_j = \mu_j + \lambda_{j1}F_1 + \lambda_{j2}F_2 + \cdots + \lambda_{jm}F_m + \epsilon_j$$

where μ_j is the mean of the variable, λ_{jk} is the loading of variable j on factor k and ϵ_j is the unique variance of variable j.

To improve interpretability, Varimax rotation was applied. This orthogonal rotation method simplifies the factor structure by maximizing the variance of squared loadings, resulting in clearer separation of variables across factors.

4.5.3 Interpretation

The number of factors is determined using Parallel Analysis. Each factor is interpreted based on the magnitude of factor loadings. High loadings indicate strong association between a pollutant and a factor. Factors are interpreted as underlying pollution sources. In this study, the factors are interpreted as Particulate pollution factor, Vehicular and secondary pollution factor, Industrial pollution factor. Factor scores are computed for each monitoring station, representing the contribution of each factor at that location. These scores are subsequently used as inputs for clustering analysis.

4.6 K-means Clustering

K-means clustering is an unsupervised learning technique used to partition a dataset into a predefined number of clusters based on similarity. The method groups observations such that intra-cluster similarity is maximized relative to inter-cluster distances.

In this study, K-means clustering is applied to the factor scores obtained from Factor Analysis in order to group monitoring stations based on their pollution profiles.

4.6.1 Objectives of Clustering

The main objectives of applying K-means clustering in this study are:

- To classify monitoring stations into groups with similar pollution characteristics
- To identify distinct pollution regimes
- To summarize pollution patterns at the state level

4.6.2 Mathematical Formulation

Let $x_1, x_2, \cdots, x_k$ represent the observations (factor scores). The aim of K-means clustering is to partition the data into K clusters $C_1, C_2, \cdots, C_k$ such that the within-cluster variation is minimized.

The objective function is given by:

$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

where μ_k is the centroid of cluster k.

4.6.3 Selection of Number of Clusters

The optimal number of clusters is determined using **Elbow Method** which examines the within-cluster sum of squares. Based on this criteria, the number of clusters is selected as $K = 3$.

5 Results

5.1 Exploratory Data Analysis (EDA)

5.1.1 Summary Statistics

Table 5.1 summarizes the summary statistics of the 7 pollutants after cleaning the data.

Table 1: Summary Statistics of Pollutant Concentrations

Pollutant	**Mean**	**Standard Deviation**	**Median**
PM2.5	110.65	67.34	92
PM10	117.07	53.83	106
NO_2	34.94	27.11	28
SO_2	16.48	13.51	12
CO	35.04	23.30	29
Ozone	50.32	37.30	41
NH_3	6.55	4.98	5

The summary statistics indicate substantial variation across pollutants, with particulate matter ($PM2.5$ and $PM10$) showing the highest mean concentrations and considerable variability across stations. The mean values exceed the medians, suggesting right-skewed distributions typical of environmental data.

Among gaseous pollutants, NO_2 shows moderate levels, while SO_2 and NH_3 exhibit relatively lower concentrations with reduced variability. Ozone and CO exhibit moderate dispersion, indicating fluctuations across locations.

Overall, particulate matter emerges as the dominant pollutant, while gaseous pollutants display comparatively lower and more stable concentrations.

5.1.2 Distribution of Pollutants

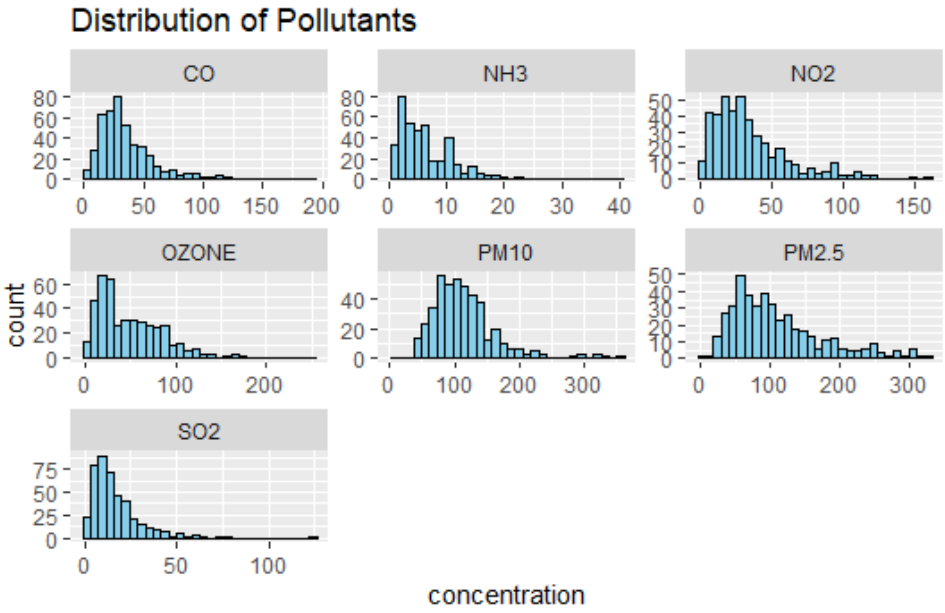


Figure 1: Distribution of Pollutants

The distributions of pollutant concentrations are generally right-skewed, indicating the presence of extreme values at certain monitoring stations. Particulate matter shows higher variability compared to gaseous pollutants, while ammonia exhibits relatively stable levels. These patterns are typical of environmental data and justify the use of data transformation and multivariate analysis techniques.

5.1.3 Correlation Analysis

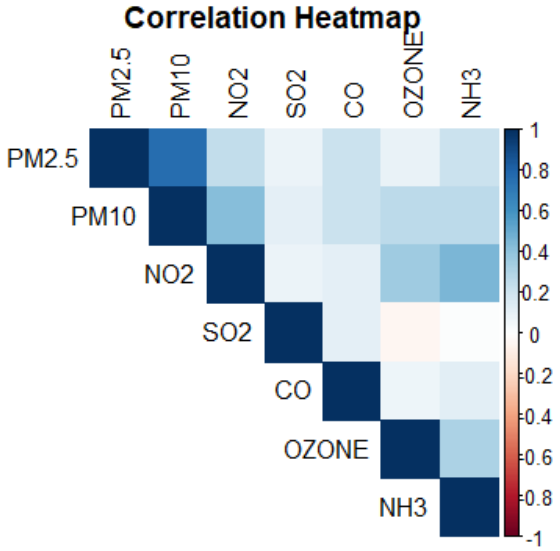


Figure 2: Correlation Heatmap

The correlation analysis reveals strong positive relationships among several pollutants, particularly between $PM2.5$ and $PM10$, indicating common emission sources. The presence of significant correlations justifies the use of multivariate techniques to uncover underlying pollution patterns.

5.1.4 Variation Across States

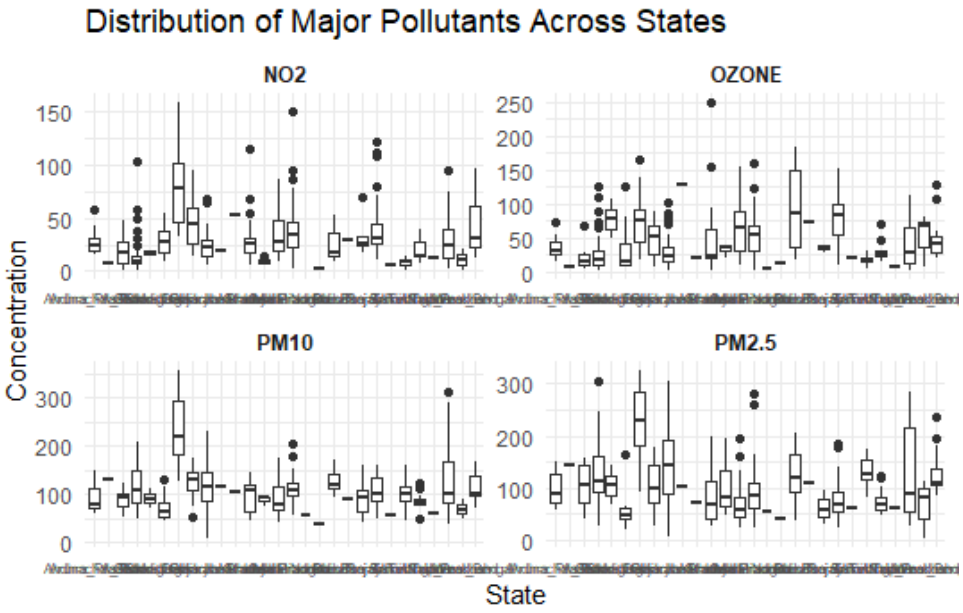


Figure 3: Distribution Across States

Boxplots of pollutant concentrations across states indicate substantial variability in pollution levels. Certain states exhibit higher median concentrations and wider spreads, suggesting uneven distribution of pollutants across regions. This observed variation motivates the use of statistical testing to formally assess differences across states.

5.2 Kruskal-Wallis Test Results

The Kruskal–Wallis test was applied to examine whether pollutant concentrations differ significantly across states.

Table 2: Kruskal–Wallis Test Results for Pollutants Across States

Pollutant	**Test Statistic (H)**	**p-value**
PM2.5	148.59	9.25×10^{-19}
PM10	144.28	5.57×10^{-18}
NO_2	153.22	5.40×10^{-20}
SO_2	93.81	2.61×10^{-9}
CO	74.41	2.58×10^{-6}
OZONE	131.65	9.96×10^{-16}
NH_3	117.55	5.97×10^{-14}

The results show that all pollutants exhibit statistically significant differences across states, with p-values well below the 0.05 significance level. This indicates that pollution levels vary significantly across regions. Among the pollutants, $PM2.5$, $PM10$, and NO_2 show particularly high test statistics, suggesting greater variability across states compared to other pollutants. These findings confirm the presence of spatial heterogeneity in pollution and justify the use of multivariate techniques to explore underlying pollution structures.

5.3 Principal Component Analysis (PCA) Results

Principal Component Analysis (PCA) was applied to the log-transformed and standardized pollutant data to examine the underlying structure and variability in air pollution across monitoring stations.

The eigenvalues obtained from PCA indicate that the first principal component (PC1) has an eigenvalue of approximately 2.47, explaining 35.34% of the total variance. The second principal component (PC2) has an eigenvalue of 1.33, contributing 19.04% of the variance. The third component (PC3) explains 14.15%, while the fourth component (PC4) contributes 12.26%. Cumulatively, the first four components account for approximately 80.79% of the total variance, indicating that a substantial portion of the variability in the data can be captured using a reduced number of components.

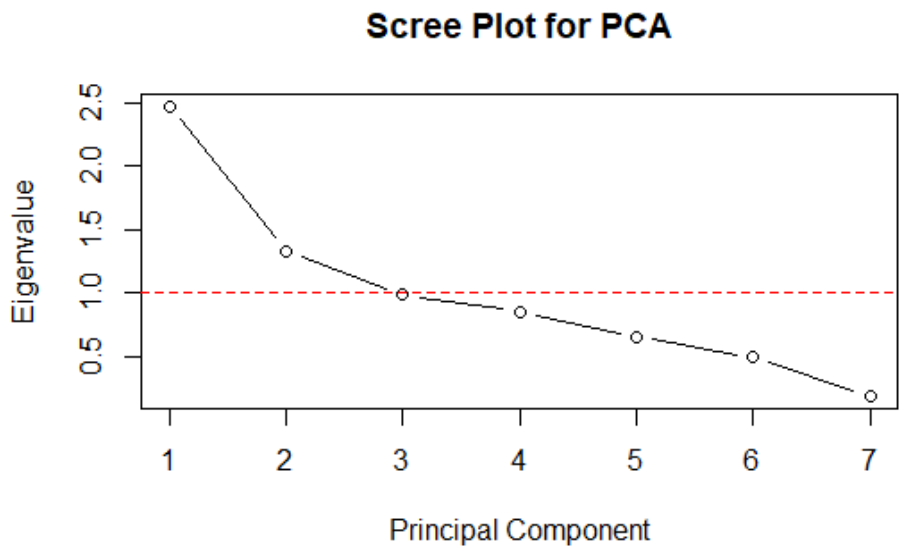


Figure 4: Scree Plot for PCA

The scree plot shows a decreasing trend in eigenvalues with a noticeable leveling off after the second or third component. Additionally, based on the Kaiser criterion (eigenvalue > 1), the first two components are clearly significant, while subsequent components contribute progressively smaller amounts of variance.

Table 3: Principal Component Loadings

Pollutant	**PC1**	**PC2**	**PC3**	**PC4**
PM2.5	0.470	0.433	-0.293	0.156
PM10	0.521	0.309	-0.289	0.137
NO_2	0.454	-0.322	-0.003	-0.173
SO_2	0.125	0.454	0.474	-0.706
CO	0.197	0.134	0.761	0.580
OZONE	0.317	-0.500	0.128	0.068
NH_3	0.383	-0.374	0.103	-0.297

The loadings indicate that the first principal component (PC1) has positive contributions from most pollutants, particularly $PM2.5$, $PM10$, and NO_2, suggesting that it represents overall pollution intensity. The second component (PC2) reflects contrasts among pollutants, with ozone and NO_2 showing opposite directions, indicating differences in pollution sources and atmospheric processes. The third and fourth components capture more specific patterns, with

CO showing strong loading on PC3 and SO_2 on PC4, suggesting localized or source-specific variations.

Overall, the PCA results suggest that the dataset can be effectively represented using a reduced number of underlying dimensions, with the first few components capturing a substantial proportion of the total variance. However, since principal components are not always directly interpretable in terms of real-world pollution sources, Factor Analysis is employed in the next step to obtain more meaningful and interpretable latent factors.

5.4 Factor Analysis (FA) Results

Factor Analysis was applied to the standardized pollutant data to identify underlying latent factors governing air pollution patterns. Prior to analysis, the suitability of the data was confirmed using the KMO measure and Bartlett's test of sphericity, indicating that the data were appropriate for factor extraction.

The KMO value 0.64 indicates acceptable sampling adequacy, and Bartlett's test is highly significant ($p < 0.001$), confirming that the data are suitable for Factor Analysis.

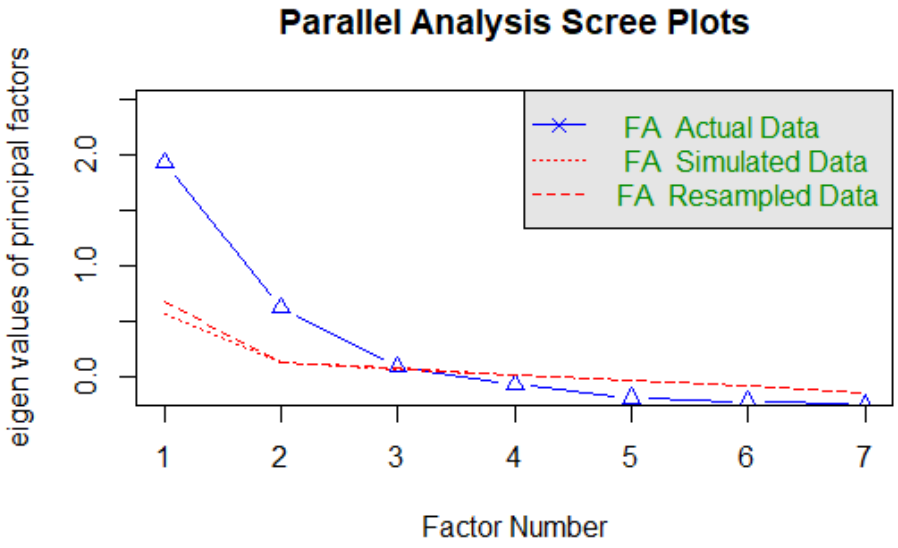


Figure 5: Parallel Analysis Plot for FA

Based on parallel analysis and scree plot criteria, three factors were retained. The model shows a good fit, with a high Tucker–Lewis Index (TLI = 1.008) and a near-zero RMSEA, indicating that the selected factor structure adequately represents the data.

The rotated factor loadings reveal distinct patterns among pollutants:

Table 4: Rotated Factor Loadings (Varimax Rotation)

Pollutant	**Factor1**	**Factor2**	**Factor3**
PM2.5	0.991		
PM10	0.775	0.292	
NO_2	0.219	0.745	
SO_2	0.100		0.991
CO	0.139	0.122	0.117
OZONE		0.540	
NH_3	0.139	0.582	

- **Factor 1 (Particulate Pollution):** This factor is strongly associated with $PM2.5$ and $PM10$, indicating that it represents particulate matter pollution, which is typically linked to combustion processes, dust, and vehicular emissions.
- **Factor 2 (Vehicular & Secondary Pollution):** This factor shows high loadings for NO_2, $OZONE$, and NH_3, suggesting a combination of vehicular emissions and secondary atmospheric processes.
- **Factor 3 (Industrial Pollution):** This factor is dominated by SO_2, which is commonly associated with industrial activities and fuel combustion.

Carbon monoxide (CO) does not load strongly on any single factor, indicating that it may be influenced by multiple sources and does not distinctly belong to a single pollution category.

The three factors together explain approximately 57% of the total variance, indicating that a substantial portion of the variability in pollutant concentrations is captured by these underlying factors.

Factor scores were computed for each monitoring station, representing the contribution of each latent factor at a given location. These scores provide a reduced-dimensional representation of pollution profiles and are used for further clustering analysis.

The results indicate that air pollution can be effectively described by three main underlying sources: particulate, vehicular/secondary, and industrial pollution. However, the presence of moderate cross-loadings and unexplained variance suggests that pollution sources are not completely distinct and may overlap across regions.

5.5 K-means Clustering Results

K-means clustering was applied to the factor scores obtained from Factor Analysis in order to group monitoring stations based on their pollution profiles. Prior to clustering, the factor scores were standardized to ensure equal weighting of each factor dimension.

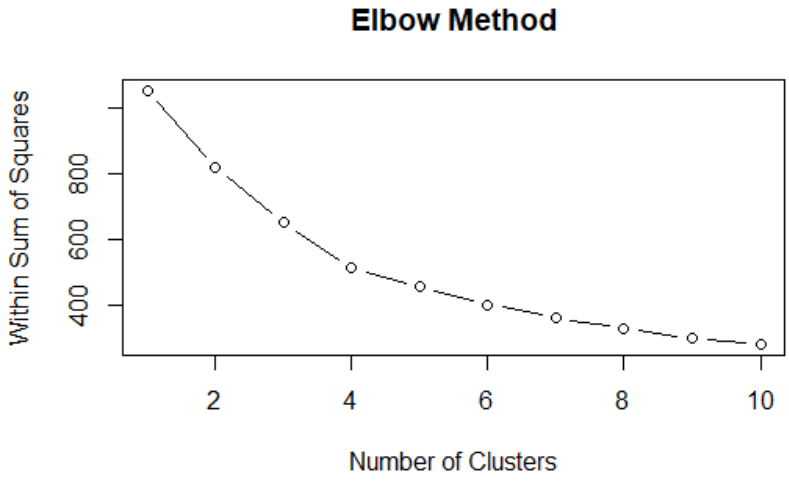


Figure 6: Elbow Method for Selecting Optimum Number of Clusters

The optimal number of clusters was determined using the elbow method. The plot of within-cluster sum of squares (WSS) showed a noticeable reduction up to K = 3, after which the rate of decrease slowed. Based on this observation and for consistency with the three-factor structure identified in Factor Analysis, the number of clusters was fixed at three.

The cluster centers (mean standardized factor scores) are as follows:

Table 5: Cluster Centers Based on Factor Scores

Cluster	**Particulate**	**Vehicular_Secondary**	**Industrial**
Cluster 1	-0.815	0.319	0.376
Cluster 2	-0.018	-1.338	-0.237
Cluster 3	0.767	0.538	-0.202

Based on the cluster center patterns, the clusters were interpreted and labeled as follows:

- **Industrial–Vehicular (Cluster 1):** Represents stations with mixed industrial and vehicular pollution but relatively low particulate matter.

- **Low Pollution (Cluster 2):** Represents stations with generally low pollution levels across all factors.
- **Particulate–Vehicular (Cluster 3):** Represents stations dominated by particulate matter and vehicular emissions, characteristic of urban environments.

The overall distribution of stations across clusters indicates that **Particulate–Vehicular** (138 stations) is the most prevalent category, followed by **Industrial–Vehicular** (128 stations), and **Low Pollution** (86 stations).

Table 6: State-wise Distribution of Stations Across Clusters (Selected States)

State	Industrial–Vehicular	Low Pollution	Particulate–Vehicular
Delhi	0	0	29
Maharashtra	25	8	20
Uttar Pradesh	15	8	14
West Bengal	2	4	15
Kerala	0	2	0
Arunachal Pradesh	0	1	0

The following plot clearly illustrates the proportion of monitoring stations belonging to each cluster across different states.

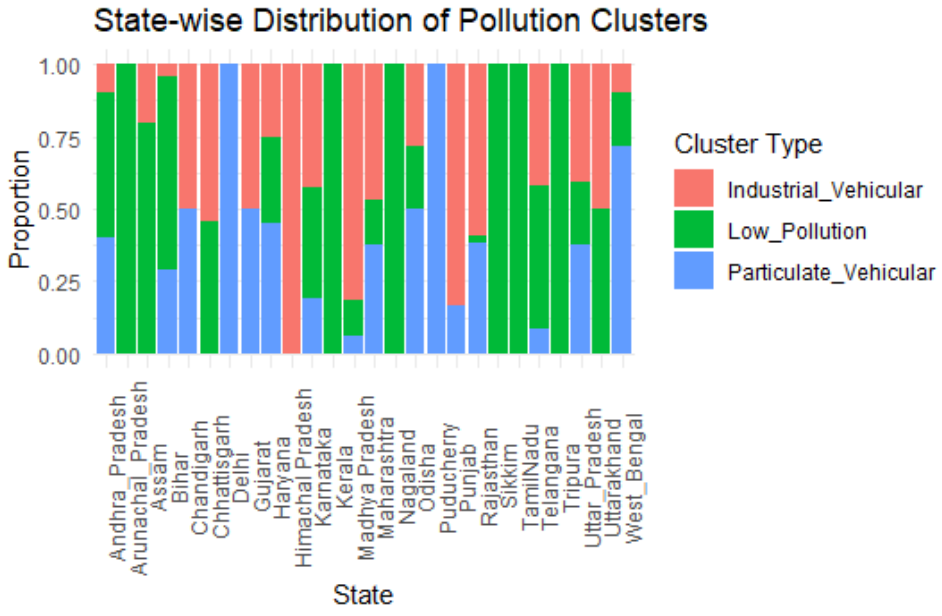


Figure 7: State-wise Distribution of Pollution Clusters

The distribution of clusters across states reveals significant regional variation in pollution patterns.

- **Urban-dominated pollution:** Delhi shows 100% Particulate–Vehicular pollution, indicating strong dominance of traffic and urban emissions. Similarly, West Bengal (approximately 71%) and Odisha (approximately 50%) exhibit high proportions of particulate–vehicular pollution.
- **Low pollution regions:** States such as Arunachal Pradesh, Nagaland, Sikkim, and Tripura show 100% Low Pollution, indicating relatively cleaner environments. Kerala and Tamil Nadu also show a strong presence of low pollution stations.
- **Industrial influence:** States such as Madhya Pradesh (approximately 81%), Punjab (approximately 83%), and Rajasthan (approximately 60%) are dominated by Industrial–Vehicular pollution, reflecting the impact of industrial activities.
- **Mixed pollution patterns:** Large and diverse states such as Maharashtra, Karnataka, Uttar Pradesh, and Telangana exhibit a combination of all three clusters, indicating the coexistence of multiple pollution sources.
- **Balanced or dual patterns:** Some regions, such as Gujarat and Chandigarh, show nearly equal proportions of Industrial–Vehicular and Particulate–Vehicular clusters, suggesting overlapping pollution regimes.

The clustering results highlight the complex and heterogeneous nature of air pollution across regions. While certain states exhibit clear dominance of specific pollution types, many states show mixed patterns due to overlapping sources such as vehicular emissions, industrial activity, and atmospheric processes. It is also important to note that states with a small number of monitoring stations (e.g., Arunachal Pradesh, Sikkim, Tripura) may show extreme proportions (e.g., 100%), which should be interpreted with caution.

The silhouette score of 0.23 confirms moderate cluster separation, consistent with the overlapping nature of pollution sources identified in the Factor Analysis. Environmental data rarely produces sharply defined clusters, as pollution regimes transition continuously across geographic regions rather than forming discrete boundaries.

Overall, K-means clustering reveals three distinct pollution regimes across monitoring stations: industrial–vehicular, low pollution, and particulate–vehicular. The results demonstrate significant spatial heterogeneity in pollution patterns and provide a meaningful classification of stations based on their dominant pollution characteristics.

6 Conclusion

This study analyzed air pollution patterns across monitoring stations in India using a combination of exploratory and multivariate statistical techniques. The results reveal substantial variability in pollutant concentrations across states, with particulate matter (PM2.5 and PM10) emerging as the dominant contributors to overall pollution load.

The Kruskal–Wallis test confirmed statistically significant differences across states, indicating strong spatial heterogeneity in air quality. Principal Component Analysis (PCA) showed that the data can be represented using a reduced number of dimensions, while Factor Analysis (FA) identified three meaningful latent factors: particulate, vehicular/secondary, and industrial pollution.

K-means clustering further grouped monitoring stations into three pollution regimes: industrial–vehicular, low pollution, and particulate–vehicular. The results highlight clear regional patterns, with urban areas such as Delhi dominated by particulate and vehicular emissions, while less industrialized regions exhibit relatively low pollution levels. Several states show mixed pollution characteristics, reflecting the coexistence of multiple sources.

Despite these insights, the study has limitations. Unequal distribution of monitoring stations across states may affect representativeness, and states with very small sample sizes may show extreme cluster proportions. Missing values were handled through filtering, and the analysis does not account for temporal variations such as seasonal fluctuations. The clustering results also indicate moderate overlap between pollution patterns.

Future work could incorporate temporal analysis, advanced clustering methods such as hierarchical clustering or DBSCAN, and additional variables like meteorological factors to improve understanding of pollution dynamics.

In conclusion, the study demonstrates that multivariate techniques provide an effective framework for analyzing complex environmental data and identifying key pollution patterns relevant to evidence-based air quality policy and regional management strategies.

7 Project Programme Code

```r
rm(list=ls())
#Loading necessary libraries
library(readxl)
library(dplyr)
library(tidyr)
library(psych)
library(ggplot2)
library(corrplot)
library(cluster)
#Loading the data
df_raw <- read_excel("C:/Users/lenovo/Desktop/M.Sc. Dissertation/pollution_data.xlsx")
# Clean invalid values
df_raw$pollutant_avg[df_raw$pollutant_avg %in% c("NA","--","")] <- NA
df_raw$pollutant_avg <- as.numeric(df_raw$pollutant_avg)
# Select relevant columns
df_selected <- df_raw %>%
  select(station, state, city, latitude, longitude, pollutant_id, pollutant_avg)
# Handle duplicates
df_selected <- df_selected %>%
  group_by(station, state, city, latitude, longitude, pollutant_id) %>%
  summarise(pollutant_avg = mean(pollutant_avg, na.rm = TRUE), .groups = "drop")
# Convert to wide format
df_wide <- df_selected %>%
  pivot_wider(
    id_cols = c(station, state),
    names_from = pollutant_id,
    values_from = pollutant_avg
  )
pollutants <- c("PM2.5","PM10","NO2","SO2","CO","OZONE","NH3")
df_clean <- df_wide %>%
  filter(rowSums(is.na(across(all_of(pollutants)))) <= 2)
df_long <- df_clean %>%
  pivot_longer(cols = all_of(pollutants),
               names_to = "pollutant",
```

```
                values_to = "concentration")
#EDA
ggplot(df_long, aes(x = concentration)) +
  geom_histogram(bins = 30) +
  facet_wrap(~ pollutant, scales = "free")
ggplot(df_long %>% filter(pollutant %in% main_pollutants),
       aes(x = state, y = concentration)) +
  geom_boxplot() +
  facet_wrap(~ pollutant, scales = "free", ncol = 2) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(size = 6),   # reduce size instead of
       rotating
    strip.text = element_text(face = "bold")
  ) +
  labs(title = "Distribution of Major Pollutants Across States",
       x = "State",
       y = "Concentration")
cor_mat <- cor(df_clean[, pollutants],
               use = "pairwise.complete.obs",
               method = "spearman")
corrplot(cor_mat,
         method = "color",
         type = "upper",
         tl.col = "black",
         title = "Correlation Heatmap",
         mar = c(0,0,1,0))
#Kruskal-Wallis Test
kw_results <- lapply(pollutants, function(p) {
  formula <- as.formula(paste(p, "~ state"))
  test <- kruskal.test(formula, data = df_clean)
  data.frame(
    pollutant = p,
    statistic = test$statistic,
    p_value = test$p.value
  )
})
kw_results_df <- do.call(rbind, kw_results)
print(kw_results_df)
```

```r
#PCA
df_pca <- df_clean %>%
  drop_na(all_of(pollutants))
station_info <- df_pca[, c("station", "state")]
pca_data <- df_pca[, pollutants]
pca_scaled <- scale(log1p(pca_data))
pca_result <- prcomp(pca_scaled, center = TRUE, scale. = TRUE)
summary(pca_result)
eigenvalues <- pca_result$sdev^2
print(eigenvalues)
plot(eigenvalues, type = "b",
     xlab = "Principal Component",
     ylab = "Eigenvalue",
     main = "Scree Plot for PCA")
abline(h = 1, col = "red", lty = 2)
var_explained <- summary(pca_result)$importance
print(var_explained)
loadings <- pca_result$rotation
round(loadings, 3)
#FA
df_fa <- df_clean %>%
  drop_na(all_of(pollutants))
fa_data <- df_fa[, pollutants]
fa_scaled <- scale(log1p(fa_data))
KMO(fa_scaled)
cortest.bartlett(fa_scaled)
fa.parallel(fa_scaled, fa = "fa", n.iter = 100)
nfactors <- 3
fa_result <- fa(fa_scaled, nfactors = nfactors, rotate = "varimax",
                fm = "ml")
print(fa_result)
loadings_fa <- fa_result$loadings
print(loadings_fa, cutoff = 0.3)
print(loadings_fa)
fa_scores <- as.data.frame(fa_result$scores)
final_fa <- cbind(df_fa[, c("station", "state")], fa_scores)
#K-means Clustering
cluster_data <- final_fa[, c("Particulate",
                             "Vehicular_Secondary",
```

```
                                "Industrial")]
cluster_scaled <- scale(cluster_data)
wss <- numeric(10)
for(i in 1:10){
  kmeans_model <- kmeans(cluster_scaled, centers = i, nstart = 25)
  wss[i] <- kmeans_model$tot.withinss
}
plot(1:10, wss, type = "b",
     xlab = "Number of Clusters",
     ylab = "Within Sum of Squares",
     main = "Elbow Method")
kmeans_result <- kmeans(cluster_scaled, centers = 3, nstart = 25)
final_fa$cluster <- as.factor(kmeans_result$cluster)
cluster_centers <- kmeans_result$centers
print(cluster_centers)
final_fa$cluster_label <- recode(final_fa$cluster,
                                 "1" = "Industrial_Vehicular",
                                 "2" = "Low_Pollution",
                                 "3" = "Particulate_Vehicular")
state_cluster_distribution <- prop.table(
  table(final_fa$state, final_fa$cluster_label),
  margin = 1
)
print(state_cluster_distribution)
state_cluster_counts <- table(final_fa$state, final_fa$cluster_label)
print(state_cluster_counts)
final_fa %>%
  count(state, cluster_label) %>%
  group_by(state) %>%
  mutate(prop = n / sum(n)) %>%
  ggplot(aes(x = state, y = prop, fill = cluster_label)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90)) +
  labs(title = "State-wise Distribution of Pollution Clusters",
       x = "State",
       y = "Proportion",
       fill = "Cluster Type")
table(final_fa$cluster_label)
```

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Appendix: Complete State-wise Distribution of Clusters

Table 7: Complete State-wise Distribution of Stations Across Clusters

State	Industrial–Vehicular	Low Pollution	Particulate–Vehicular
Andhra Pradesh	1	5	4
Arunachal Pradesh	0	1	0
Assam	1	4	0
Bihar	1	16	7
Chandigarh	1	0	1
Chhattisgarh	6	5	0
Delhi	0	0	29
Gujarat	8	0	8
Haryana	5	6	9
Himachal Pradesh	1	0	0
Karnataka	9	8	4
Kerala	0	2	0
Madhya Pradesh	13	2	1
Maharashtra	25	8	20
Nagaland	0	1	0
Odisha	4	3	7
Puducherry	0	0	1
Punjab	5	0	1
Rajasthan	25	1	16
Sikkim	0	1	0
Tamil Nadu	0	3	0
Telangana	5	6	1
Tripura	0	1	0
Uttar Pradesh	15	8	14
Uttarakhand	1	1	0
West Bengal	2	4	15